Witchayut Tangwongkitsiri

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Portfolio: https://knotnot.github.io/portfolio/

SUMMARY

Bachelor's degree in electrical engineering, *Chulalongkorn University* (2021 - 2025)

Electrical Engineering graduate from Chulalongkorn University with hands-on experience building end-to-end data pipelines across personal and internship projects. Focused on core data engineering principles — modeling, partitioning, and scalable architecture over tool familiarity alone.

EXPERIENCE

IOT Development Intern

Electrical Engineering Technology for Agricultural Resources (EETAR) (June 2024 – August 2024)

- Designed and implemented an **end-to-end IoT data pipeline (sensor → database → visualization)** across 3 sensor nodes, each equipped with 6 sensors measuring soil moisture and plant growth.
- **Collected, cleaned, and stored time-series agricultural data into InfluxDB** every 30 minutes, accumulating 36,000+ records over 6 weeks for comparative analysis

Project

Design of Software-Defined Wireless Network for Machine Learning-Driven DDoS Detection and Tiered Response. (IEEE ISCIT 2025, Co-author)

- Developed a load balancer for server, Built a server-side log ingestion pipeline that collects web traffic logs from VM-hosted servers and batch-loads them into **MySQL** every 5 seconds for near-real-time analysis.
- Engineered structured datasets from normal and DDoS traffic to enable near-real-time machine learning–based anomaly detection and interactive traffic visualization dashboards in **Power BI.**

Build Data Pipeline ETL & Automation (TripAdvisor API)

- Developed a Python-based ETL pipeline to extract, clean, transform, and load TripAdvisor data into a Dockerized **MySQL database.**
- Designed and implemented DAGs in **Apache Airflow,** managing task dependencies, retries, and scheduling to ensure robust **and automated** daily data ingestion.

Flask-based Machine Learning Web Application

- Developed a **ML pipeline** by preprocessing and merging diverse fertilizer and weather datasets, focusing on feature engineering and normalization.
- Deployed the model via a **Flask web application**, providing a user-friendly interface for real-time predictions and practical accessibility.

Automated Multi-Source Data Pipeline (GCP)

- Built an automated pipeline using **Python** and **Cloud Run** to ingest data via APIs and Web Scraping into **GCS** every 5 minutes.
- Orchestrated 3 concurrent tasks via **Cloud Scheduler,** transforming raw data into structured **BigQuery** tables with optimized Schema design for analytical use.

ADDITIONAL & SKILLS

- **Language:** Thai (Native), English (Fluent), TOEIC score: 805/990
- **Cloud & Data Platforms:** Google Cloud Platform, AWS, GCS, InfluxDB, MySQL, PostgreSQL
- **Data Engineering:** End-to-End ETL/ELT Pipelines, Data Transformation, Regex, Web Scraping, Data modeling
- **Programming & Tools:** Python, SQL, Apache Airflow, Git, Power BI, Ubuntu, Docker
- **Soft Skill:** Fast Learner, Effective Communication, Collaborative Team Player, Problem-Solving Mindset